

Is Twitter for the Birds?

Estimating the causal effect of Twitter / X sentiment on stock returns across the S&P 500 and Russell 3000.

Why does Twitter sentiment matter for financial markets?

- ▶ **Social media is an information channel** — Twitter/X, Reddit, and StockTwits often double as business intelligence for retail traders.
- ▶ **Retail trading participation has doubled** — Share of U.S. equity volume from retail traders roughly doubled between 2010 and 2020.
- ▶ **Retail investment concentrated in small/mid-cap** — Retail participation is highest exactly where institutional analyst coverage is thinnest.
- ▶ **Finding market predictors** — Sentiment logically reflects in pricing. Determining to what extent it influences pricing guides real trading strategies.



Prior studies

Study	Method	Sample	Key Finding
Tetlock (2007)	OLS	WSJ column	Negative word fraction predicts Dow Jones returns; reverses within a week
Bollen et al. (2011)	GPOMS mood	Twitter, 2008	Twitter mood states improve DJIA next-day forecast accuracy
Antweiler & Frank (04)	Probit / VAR	Yahoo msg boards	Message board activity predicts next-day volatility and returns
★ Gu & Kurov (2020)	Fama-MacBeth	Twitter 2015–2017	Lagged Twitter sentiment predicts open-to-open returns at 1-day lag
★ Teti et al. (2019)	OLS panel	30 DJIA firms	Negative tweet counts predict returns at short horizons

GAP THIS PAPER FILLS

Prior studies do not test for reverse causality or examine cap-group heterogeneity across the full market-cap spectrum. Both gaps are central to this paper.

Does Twitter drive returns?

1

Does Twitter sentiment causally affect same-day intraday stock returns?

Can I show that Twitter sentiment causally affects trading as reflected in prices?

2

Can I use modern machine learning approaches to refine previous findings?

Will double debiased ML provide more accurate estimators than previous studies?

CENTRAL HYPOTHESIS

Twitter sentiment has a small but measurable impact on asset prices due to its influence on trader behavior. This impact will dissipate more quickly than other market signals.

The data

S&P 500 (recent observations)

~ 160K observations

Jan 2025 – Mar 2026

Full confounder set: FE-OLS & DoubleML

S&P 500 (extended panel)

~ 1.19M observations

Jan 2016 – Feb 2026

Matched to Fama-Frech volatility data

Russell 3000

~ 1.97M observations

Jan 2016 – Feb 2026

Replicates Gu& Kurov study

Key Variables

twitter_sent	Daily avg. Twitter sentiment [-1,+1]
news_sent	News daily avg. sentiment [-1,+1]
twitter_neg_count	Count of negative-polarity tweets
return	Open-to-close dollar return
rsi_30	30-day Relative Strength Index
mkt_cap	Market capitalization (millions)
vol_rs	Price volatility

Data Limitations

- ▶ **Sentiment black box** — Bloomberg NLP architecture and tweet filtering criteria undisclosed.
- ▶ **Temporal mismatch** — Twitter posts are released 24/7, and may influence pricing in between trading sessions.
- ▶ **Fama-French data** — Daily observations through February 2026. Captures price volatility and momentum across time.
- ▶ **Uneven Twitter data** — Bloomberg did not track twitter data for all firms (or the firms did not have accounts) for all periods.

Identification strategy

Demonstrate that sentiment changes price, but the effect decays rapidly

INITIAL ASSESSMENT

Firm Fixed-Effects OLS

- Basic linear model with confounders
- Sentiment as input variable
- Return as output variable
- Negative tweets to validate direction
- Lags (1, 2, 3, 5, 7)
- Firm-level FEs

Package: pyfixest

ML ANALYSIS

Double ML — PLR

- Address non-linear confounding
- Lags (1, 2, 3, 5, 7) via CATEs
- No reversal test
- XGBoost nuisance learner for outcome and treatment bias modeling
- 5-fold cross-fitting
- 20 rep Monte Carlo
- 1,000 bootstrap draws.

Package: DoubleMLPLR · XGBoost

VALIDATE METHODOLOGY

Gu & Kurov Study Replication

- Use previous study methodology to test identification
- Daily cross-sectional regressions averaged over time.
- Fama – French 4-factor risk-adjusted returns

Replicates Gu & Kurov (2020)

Three identification attempts

FIRST PASS

S&P 500

- FE-OLS for simplicity
- DoubleML for thoroughness
- Robustness checks for reverse causality
- Gu & Kurov replication fails
- Confusing results

SECOND ATTEMPT

Extended Panel S&P 500

- Fama-MacBeth OLS (Gu&Kurov)
- Robustness checks for reverse causality
- Interesting observations
- No significant results

FINALLY

Extended Panel Russel 3000

- Exact Gu & Kurov replication with longer panel
- Statistically significant findings
- Market cap analysis reveals possible reason for failure of S&P 500 studies

WHAT HAPPENED

Twitter sentiment appeared to play little to no role in the large, S&P 500 companies. As the sample expanded to include lesser-known firms, Twitter's impact on asset pricing grew.

S&P 500: Twitter sentiment insignificant across all FE-OLS specs

Model	Treatment	Coeff.	SE	R ²
1 — Baseline	twitter_sent	-0.370	(0.230)	0.002
2 — Full ctrls	twitter_sent	-0.422	(0.325)	0.019
3 — Neg. clip	neg_twitter_sent	-0.072	(0.271)	0.002
4 — Tweet count	twitter_neg_count	+0.001	(0.001)	0.002

Significant controls in Model 2:

2	news_sent	-1.056***	(0.114)
2	rsi_30	+0.084***	(0.016)

Key Takeaways

- ▶ **Twitter -> zero explanatory power** — $R^2=0.002$ in baseline; coefficient insignificant across all four models.
- ▶ **News sentiment works** — Coefficient of 1.056*** is 2.5× larger than any Twitter estimator.
- ▶ **RSI momentum effect** — Firms on 30-day upward trajectories continue gaining intraday.
- ▶ **Tweet volume ≠ bearish** — Positive sign on neg tweet count suggests any press may be good press, but trivial.

TAKEAWAY

In a linear OLS framework, Twitter sentiment has **no statistically detectable effect on S&P 500 intraday returns.**

DoubleML: significantly puzzling effects

Run 1 · `twitter_sent ~ return`

-0.924***

SE = 0.226 · $p < 0.01$

Higher Twitter sentiment → lower same-day return.

A 1 SD increase (~ 0.165) $\approx -0.15\text{¢}$ same-day impact.

Run 2 · `twitter_neg_count ~ return`

+0.004***

SE = 0.0015 · $p < 0.01$

More negative tweets → higher return.

A surge in negative tweets signals elevated attention not bearish info.

SIGN REVERSAL

Difference in signs indicates that Tweet volume and tweet sentiment are not necessarily connected and a surge in attention increases negative counts without tipping price downward.

News sentiment: the stronger, more reliable channel

news_sent → return (DoubleML PLR)

−0.759***

SE = 0.135 · p < 0.01 · Consistent with FE-OLS estimate (−1.056***)

Model	news_sent coeff.	Sig.
FE-OLS (Model 2)	−1.056	***
DoubleML PLR	−0.759	***

Why News Outperforms Twitter

- ▶ **Better timing alignment** — News articles have timestamps that better separate pre- and post-market content, reducing the aggregation mismatch problem.
- ▶ **Lower noise-to-signal** — Produced by professional journalists rather than the broad public; more informative per unit sentiment.
- ▶ **Convergence across methods** — OLS (−1.056) and DoubleML (−0.759) both significant; consistency suggests linear OLS approximation works for the news channel.

IMPLICATION

News captures more of the information used by traders focused on the S&P 500, making it a more useful barometer for asset performance.

Placebo testing for reverse causality

Placebo Design

Regress yesterday's return on today's Twitter sentiment within firms.
If there is no reverse causality, the coefficient should be zero.

PLACEBO RESULT

+2.348***

SE = 0.225 · $p < 0.01$ | 2.5× larger than the forward estimate (−0.924)
in absolute value, and in the wrong direction.

What This Means

- ▶ **Prices drive sentiment** — Yesterday's strong returns predict today's higher Twitter sentiment, not the other way around.
- ▶ **Time mismatch** — Bloomberg's 24-hour average captures after-hours tweets that respond to the day's already-recorded price moves.

INTERPRETATION

Asset prices appear to drive twitter sentiment, and not the other way around.

No decay pattern: lagged sentiment significant at all horizons

Lag Decay — DoubleML PLR

Expected under causal view: decay from Lag 1 → 7.

Actual: all lags significant, no decay.

twitter_sent	1	-1.215***
twitter_sent	2	-0.957***
twitter_sent	3	-1.122***
twitter_sent	5	-1.126***
twitter_sent	7	-0.674***

Impact Persistence — Current Sent → Future Return

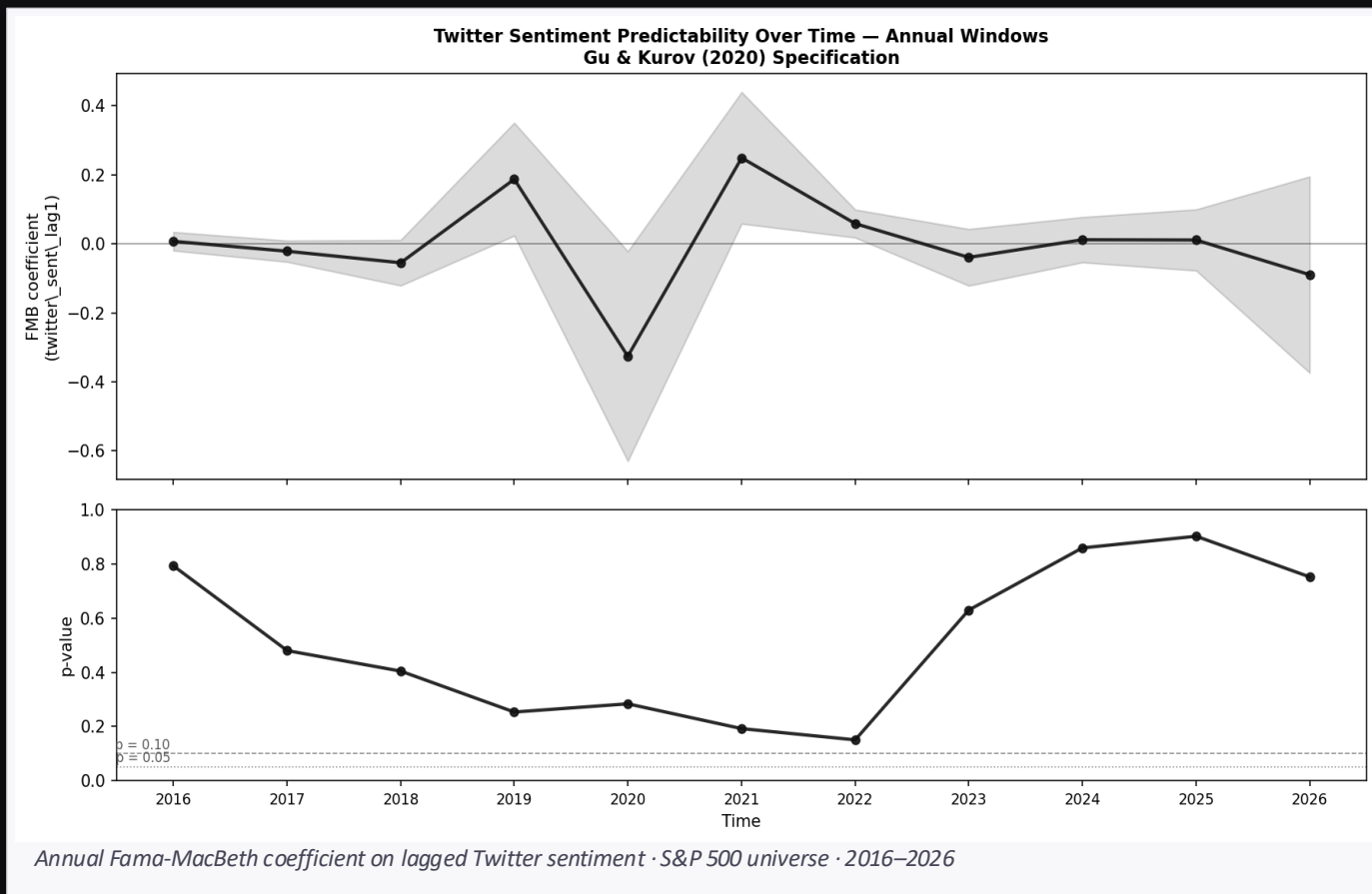
Twitter insignificant at every lead. News also shows inconsistent signs.

twitter_sent	1	-0.111	n.s.
twitter_sent	2	+0.217	n.s.
twitter_sent	7	+0.028	n.s.
news_sent	1	-0.357	**
news_sent	2	+0.312	***
news_sent	7	+0.342	***

TAKEAWAY

Because we observe no decay, reverse causality is the more likely reason for lagged sentiment–return correlations.

Long term S&P 500 analysis (2016–2026)



Key Observations

- ▶ **Insignificant every year** — The estimator (FMB coefficient) is not statistically significant in any calendar year from 2016 to 2065 for the S&P 500.
- ▶ **No pattern change** — No identifiable date where Twitter's informational role notably changed
- ▶ **Discrepancy with Gu & Kurov (2020)** — Their significant finding likely reflects a broader set of companies including smaller-cap firms.
- ▶ **Platform changes** — Decrease in p-value starting in 2022 suggest possible change in perceived trustworthiness of Twitter as a news source.

TAKEAWAY

For large-cap S&P 500 firms, Twitter sentiment has never been a reliable return predictor

Expanding to the full market-cap spectrum

Full Russell 3000 Panel OLS

+0.708***

Positive and significant

Direct contrast to S&P 500 negative (insignificant) estimate

Firm-days by cap group

763K / 641K / 568K

Large (>\$10B) · Mid (\$2B–\$10B) · Small (<\$2B)

ANALYSIS

Twitter is most informative where formal sources are sparse.
Fewer analysts, wider bid-ask spreads, more retail investors.

Cap-Group Gradient

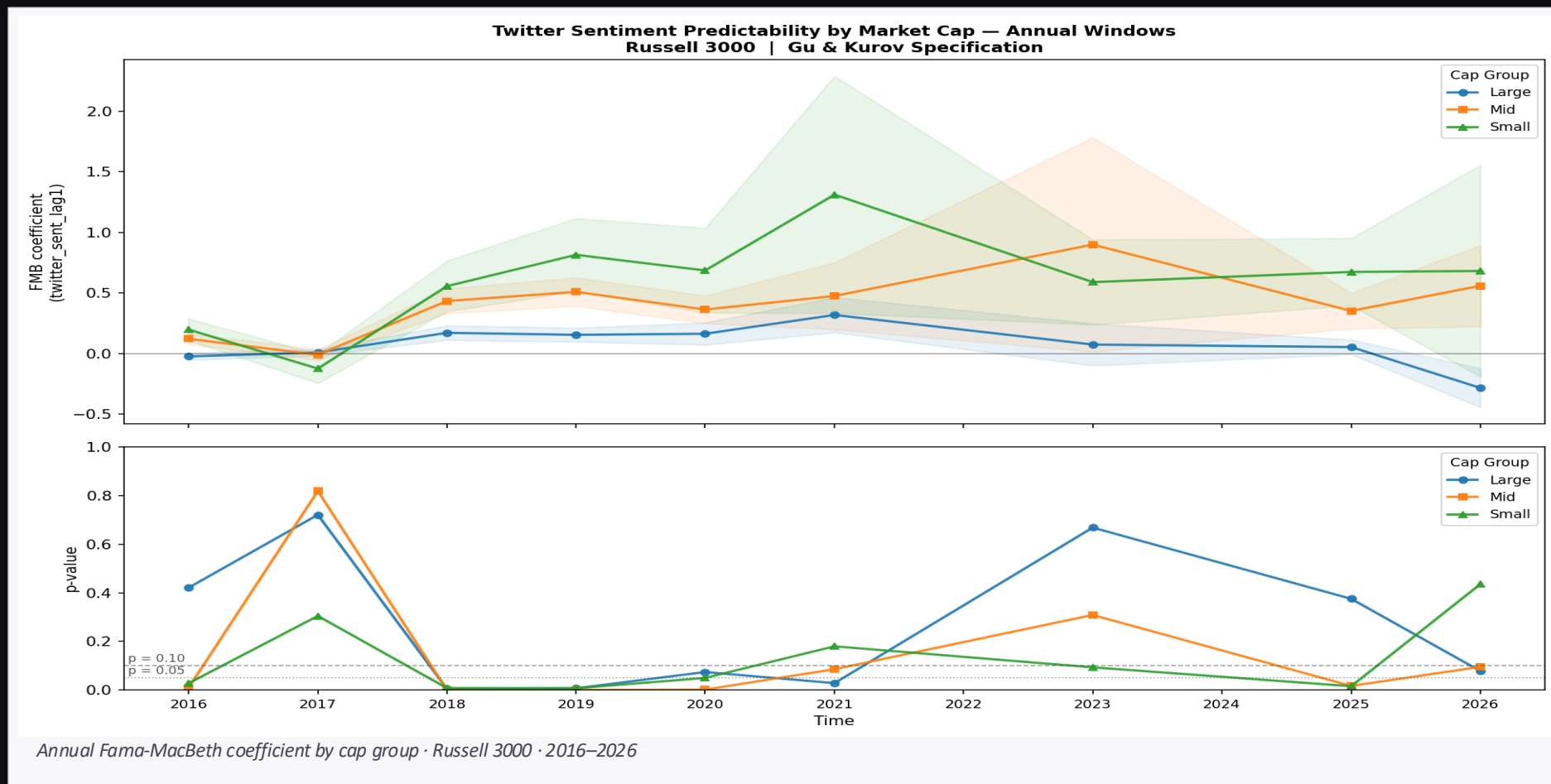


INTERACTION TERM

$twitter_sent \times log_mkt_cap = -0.148^*$

Effect is linearly decreasing with firm size

Cap-group effects consistent across time



■ Small cap stocks significant in most years; strongest 2018–2020.

■ Mid cap resembles small cap but smaller effect.

■ Large cap stocks are generally flat and insignificant.

Sentiment as a sorting variable: positive returns in 9 of 10 years

2.04

Sharpe Ratio (2023)

Peak annual Sharpe for S&P 500 long-short strategy

1.61

Sharpe Ratio (2025)

Despite insignificant FMB coefficient in that year

Strategy Design

- ▶ **Long** (buy) top-decile sentiment firms
- ▶ **Short** (sell) bottom-decile sentiment firms
- ▶ **Daily rebalance** within each cap group

How is this different and why care?

- ▶ **FMB gives and estimator** — The mean relationship between sentiment and return across all firms.
- ▶ **Sharpe Ratio exploits the tails** — Where the sentiment/return relationship may be nonlinear and stronger.
- ▶ **Provides a possible strategy** — Shows that a viable strategy can still be extracted from an insignificant result

IMPLICATION

Sentiment rankings, not sentiment levels, may be an informative signal of momentum. Worth exploring as an instrument for portfolio structure.

Interpretations of findings

THE POWER OF TWEETS

Real forward effect, but with caveats

- Twitter sentiment contains real information that is useful.
- Twitter is not used when other information sources are available.
- As company size grows Tweet-effect diminishes.

THE APPLICATION

Sentiment as a sorting signal

- Sentiment seems useful as a tool for long-short strategy.
- Expands low-grade market intelligence for small firms.
- Potential application as a momentum indicator/instrument

THE LIMITATIONS

Isolation of sentiment effect is really hard

- Twitter is 24/7, but trading is not.
- Available tools make temporal isolation difficult, weakening causal identification.
- Wide variance in tweet content makes endogeneity difficult to address.

TAKEAWAY

The gradient (+0.375 → +0.804 → +0.942 (large to small cap) indicates that Twitter is informative where formal information is sparse, and its usefulness increases as formal information decreases.

Key findings & future directions

Summary of Findings

S&P 500 FE-OLS	Twitter insignificant in all 4 specs; news_sent dominant (-1.056^{***})
S&P 500 DoubleML	-0.924^{***} but placebo $+2.348^{***}$ — reverse causality undermines causal claim
S&P 500 FMB	Insignificant every year 2016–2025; no structural break
Russell 3000 ★	$+0.708^{***}$ overall; gradient: $+0.375$ (large) $\rightarrow$ $+0.804$ (mid) $\rightarrow$ $+0.942^{***}$ (small)
Long-short	Positive in 9 of 10 years; Sharpe up to 2.04 — tails contain exploitable signal

Future Directions

- ▶ **Timestamped tweet data** — Build pre-market / intraday / after-hours sentiment windows to resolve aggregation mismatch and cleanly identify causal window.
- ▶ **DoubleML by cap group** — Apply causal estimator within each Russell 3000 cap segment to assess if identification is stronger for small-caps.
- ▶ **Multi-platform expansion** — Reddit (WallStreetBets), StockTwits, LinkedIn — assess platform-specific information channels.
- ▶ **Custom entity-level NLP** — Build custom tool to analyze tweet influence based on author, time, and/or ticker.

TAKEAWAY

Twitter sentiment is most useful where formal analysis is thinnest. For large-caps, news is the better signal.

REFERENCE MATERIAL

- A1 — Summary Statistics: S&P 500 Extended Panel (N = 1,190,286)
- A2 — Summary Statistics: Russell 3000 Panel (N = 4,445,236)
- A3 — Full FE-OLS Regression Table (4 models × all controls)
- A4 — Russell 3000 Cap-Group OLS & DoubleML All Specifications
- A5 — Gu & Kurov (2020) Replication Results
- A6 — Unused Temporal Figures: GK Monthly & Quarterly
- A7 — Unused Temporal Figures: Russell Monthly, Quarterly, Cap-Group Quarterly
- A8 — Full Variable Definitions
- A9 — DoubleML: How It Works (PLR, Cross-Fitting, XGBoost)
- A10 — Panel FE-OLS Estimating Equation & Notation
- A11 — Fama-MacBeth Equation & Fama-French 4-Factor Risk Adjustment
- A12 — Ken French Data Library — Factor Data Description
- A13 — Bloomberg Sentiment Tool & Academic Validation
- A14 — Long-Short Strategy & Sharpe Ratio — Construction & Results
- A15 — Lagged DoubleML PLR — Sentiment Persistence Across Horizons (CATE by Lag)
- A16 — Time-Value of Twitter Sentiment — What the Lag Structure Reveals

Summary Statistics — S&P 500 Extended Panel (2016–2026)

N = 1,190,286 firm-day observations · 503 S&P 500 firms · Daily frequency

Variable	Mean	Std Dev	Min	Max	N (non-missing)
return_oo (%)	0.0665	2.1386	−47.75	107.75	1,190,286
twitter_sent [−1,+1]	0.0257	0.1651	−1.000	1.000	908,781
news_sent [−1,+1]	0.0108	0.3019	−1.000	1.000	935,145
vol_rs (Rogers-Satchell)	0.0292	0.1008	0.000	21.259	1,190,789
abnorm_vol (%)	0.000	84.871	−100.0	15263.2	1,190,785
log_mkt_cap	10.363	1.118	6.218	15.431	1,190,699
bid_ask_spread (\$)	0.199	5.947	0.000	3209.6	1,190,731
return_oo_adj (%)	0.000	2.132	−47.99	107.47	1,156,294

Notes: return_oo = open-to-open return (%). vol_rs = Rogers-Satchell realized volatility. abnorm_vol = % deviation from ticker mean. return_oo_adj = Fama-French-Carhart 4-factor residual. Twitter sentiment coverage 76.3% of firm-days.

Summary Statistics — Russell 3000 Panel (2016–2026)

N = 4,445,236 firm-day observations · ~3,000 Russell 3000 firms · Daily frequency

Variable	Mean	Std Dev	Min	Max	N (non-missing)
return_oo (%)	0.0960	8.2444	−100.0	13782.4	4,445,236
twitter_sent [−1,+1]	0.0280	0.1690	−1.000	1.000	1,976,749
news_sent [−1,+1]	0.0230	0.2717	−1.000	1.000	1,924,283
vol_rs (Rogers-Satchell)	0.1064	0.7942	0.000	748.88	4,448,157
abnorm_vol (%)	0.000	152.711	−100.0	36947.0	4,447,963
log_mkt_cap	7.769	1.882	−0.933	15.431	4,444,659
bid_ask_spread (\$)	3,437.6	406,302.5	0.000	1.87×10 ⁸	4,437,473
return_oo_adj (%)	0.000	8.167	−100.5	13732.4	4,445,178

Notes: Russell 3000 panel shows wider return distribution than S&P 500 — small-cap volatility. Bid-ask spread extreme outliers driven by illiquid small-cap penny stocks. Cap groups: Large (>\$10B): 763K obs; Mid (\$2B–\$10B): 641K; Small (<\$2B): 568K.

Fixed-Effects OLS — Full Regression Table (S&P 500 Primary Panel)

Variable	M1 Baseline	M2 Full Controls	M3 Neg Clip	M4 Tweet Count
twitter_sent	-0.370 (0.230)	-0.422 (0.325)		
neg_twitter_sent			-0.072 (0.072)	
twitter_neg_count		0.00085 (0.001)		0.000533 (0.000971)
px_high		-0.085 (0.121)		
px_low		0.091 (0.122)		
mkt_cap		0.000 (0.000001)		
total_equity		0.000001 (0.000004)		
debt_to_equity		-0.000089 (0.000049)		
volume		0.000 (0.000)		
news_sent		-1.056*** (0.114)		
rsi_30		0.084*** (0.016)		
ma_50		-0.013 (0.011)		
Firm FE	✓	✓	✓	✓
Observations	160,103	147,138	160,103	150,333

Russell 3000 Cap-Group OLS & DoubleML PLR Results

Panel OLS: twitter_sent × Market Cap — Russell 3000					
	(1) Full	(2) Full+Int.	(3) Large	(4) Mid	(5) Small
twitter_sent	0.708*** (0.057)	1.941*** (0.743)	0.375*** (0.023)	0.804*** (0.036)	0.942*** (0.185)
log_mkt_cap	−0.050 (0.057)	−0.047 (0.058)	−0.048*** (0.009)	−0.009 (0.014)	−0.125 (0.162)
twit × log_mkt_cap	—	−0.148* (0.082)	—	—	—
Firm FE	Yes	Yes	Yes	Yes	Yes
Obs.	1,972,274	1,972,274	763,274	640,932	568,068
R ² (within)	0.001	0.001	0.001	0.003	0.002
Clustered SEs (entity). Dep var: return_oo_adj. *** p<0.01, ** p<0.05, * p<0.1.					

DoubleML PLR — All Specifications		
TREATMENT → RETURN		Coeff.
twitter_sent	(1)	−0.924*** (0.226)
twitter_neg_count	(2)	+0.004*** (0.0015)
news_sent	(3)	−0.759*** (0.135)
TREATMENT → px_high		
twitter_sent	(4)	+1.610*** (0.217)
twitter_neg_count	(5)	−0.002* (0.0012)
PLACEBO (sent → lag1 ret.)	(6)	+2.348*** (0.225)
XGBoost, 5-fold, 20 reps, 1K bootstrap. Placebo: lag1_return ~ twitter_sent.		

Gu & Kurov (2020) Replication — Fama-MacBeth Results

Fama-MacBeth coefficient on lagged Twitter sentiment across S&P 500 universe, 2016–2026

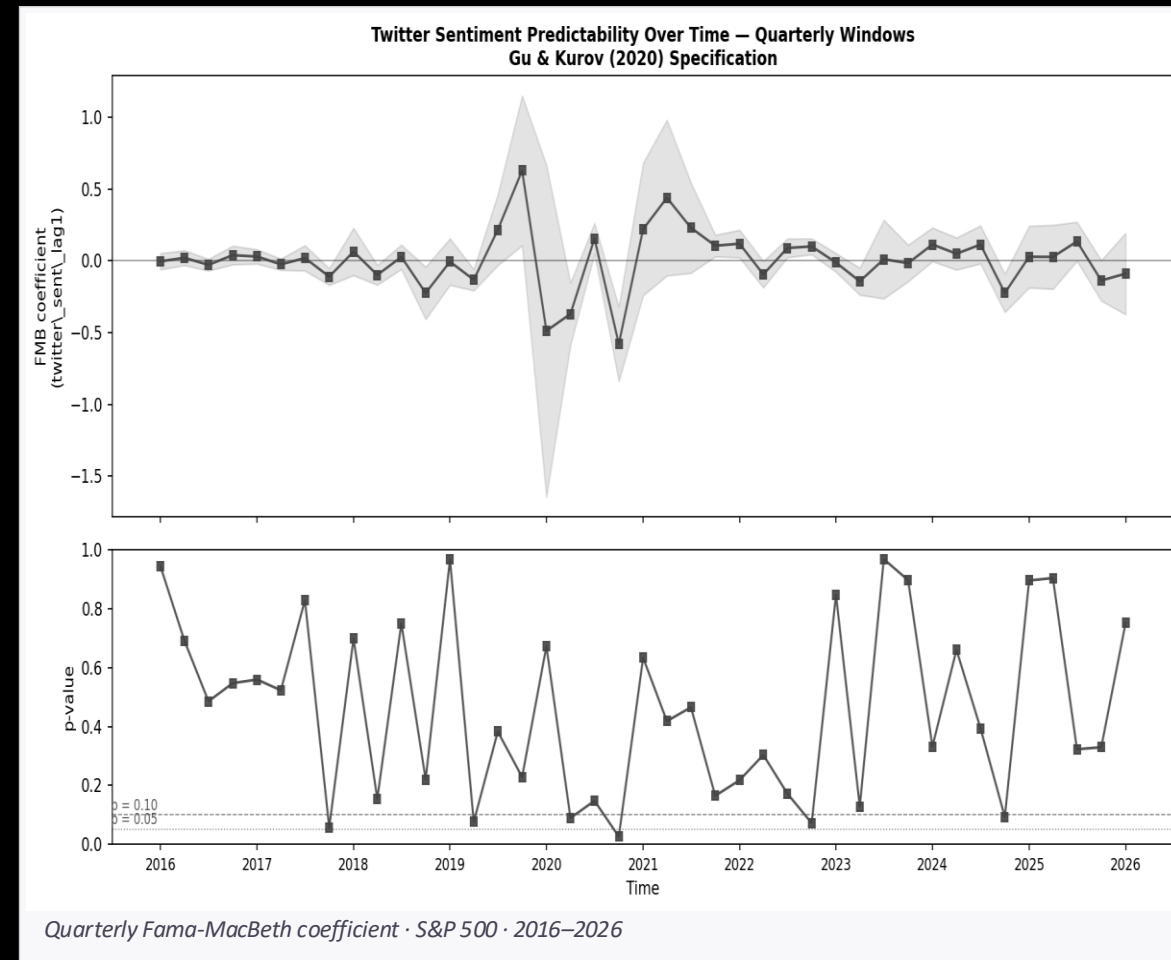
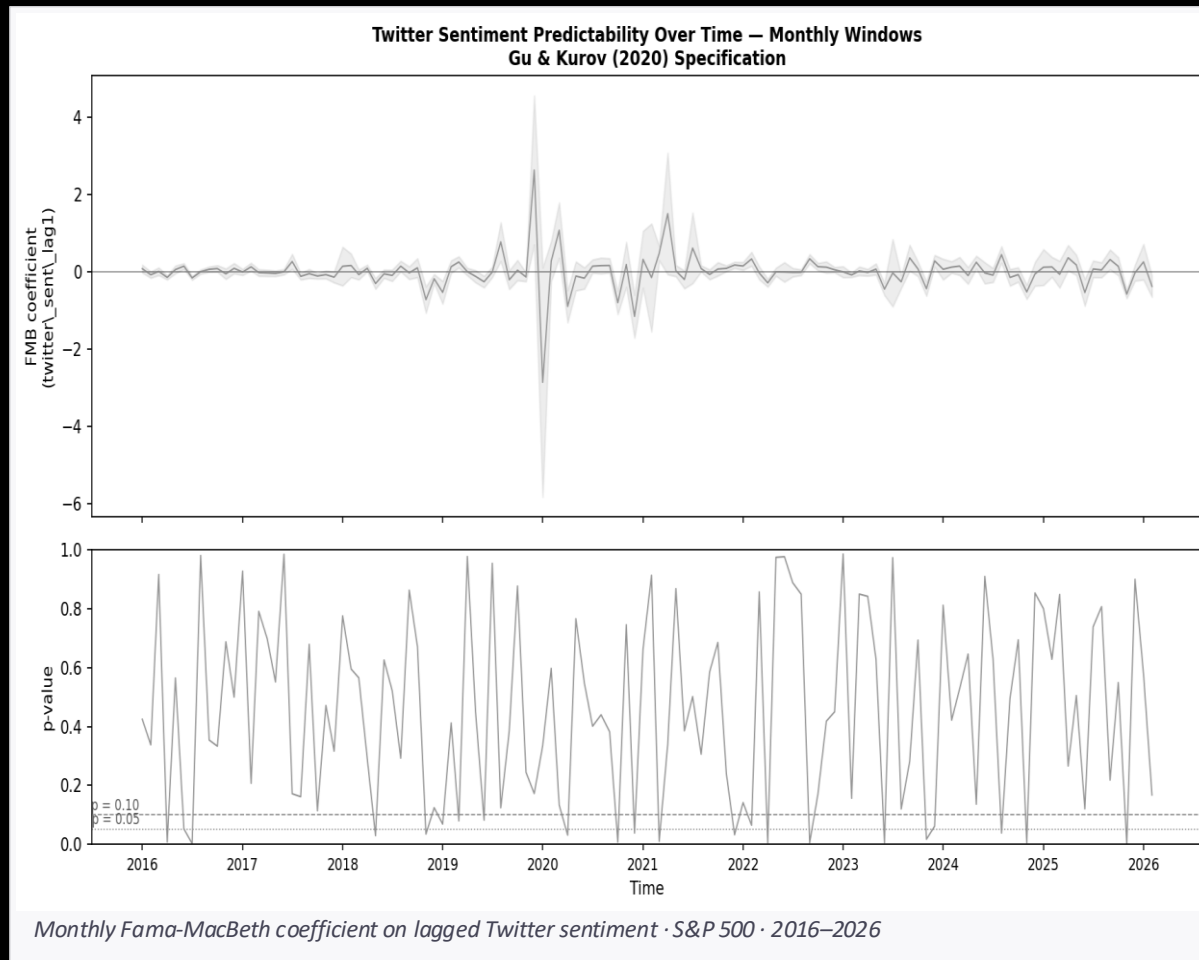
Specification	Est.	Raw Return (1)	Risk-Adj (2)	Significance	Notes
Full sample 2016–2026	FMB	Not run	−0.036 (0.049)	n.s.	Replicates Gu & Kurov Table 2; uses open-to-open FF4-adjusted return
Annual coefficients	FMB	—	All insignificant	n.s. all years	No year shows $p < 0.10$ for <code>twitter_sent_lag1</code> in S&P 500
Gu & Kurov original	FMB	—	+0.136***	***	2015–2017 sample; their universe includes smaller firms
Discrepancy explanation	—	—	—	—	Universe difference (S&P 500 here vs. broader Russell-like in G&K) + possible provider/vintage difference

Controls: 5 lags each of `return_oo_adj`, `abnorm_vol`, `vol_rs`, `log_mkt_cap`, `bid_ask_spread`. Newey-West SE (5 lags). Days with <30 firms reporting non-missing `twitter_sent` excluded.

KEY FINDING

The Gu & Kurov (2020) coefficient of +0.136*** does not replicate in the S&P 500 universe for 2016–2026. The most likely explanation is universe composition: G&K's sample appears to include smaller-cap stocks where the sentiment effect is much stronger (see Russell 3000 cap-group results).

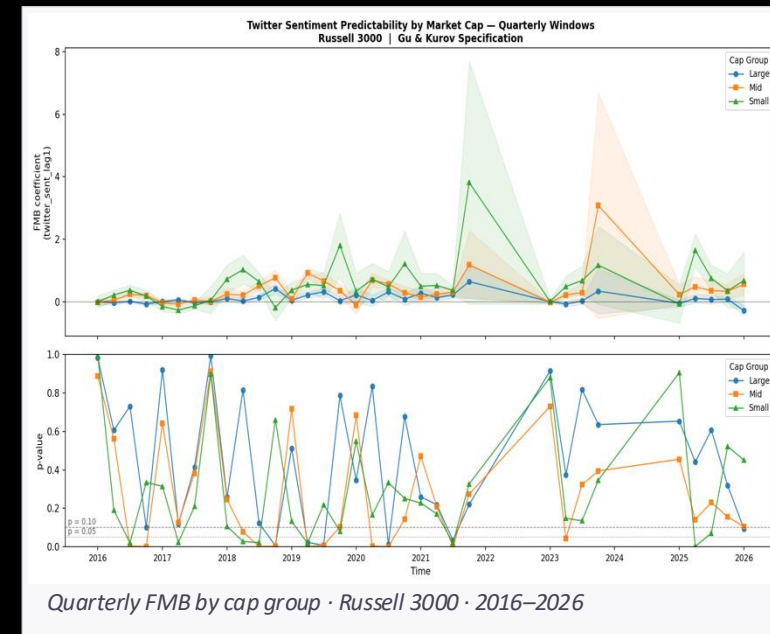
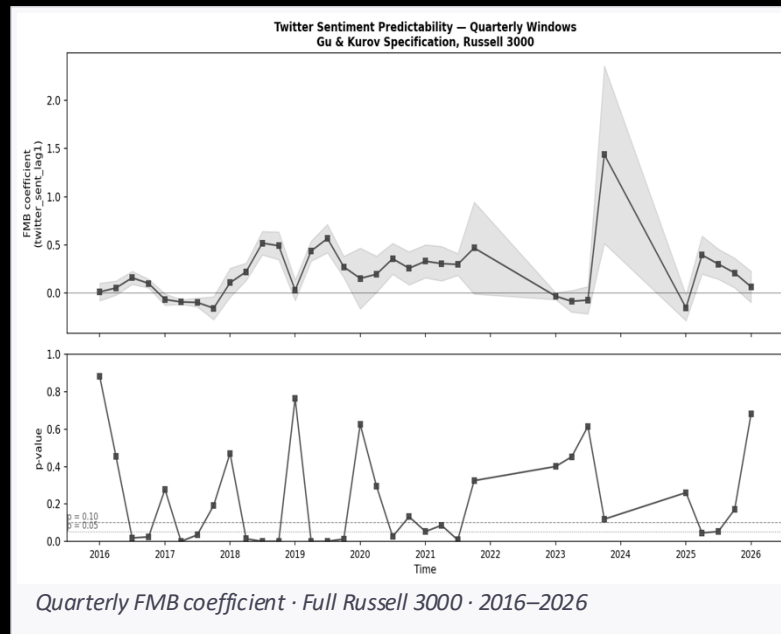
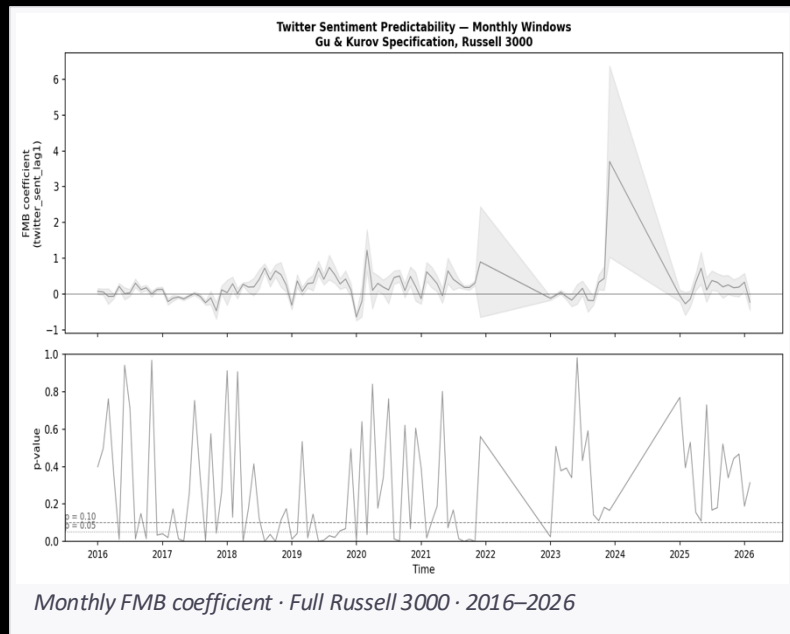
Temporal Analysis — Monthly & Quarterly FMB Coefficients (S&P 500)



INTERPRETATION

Monthly estimates are less reliable due to small cross-sections in some months. Both series confirm no persistent positive coefficients for the S&P 500. Some months show significant negative coefficients. consistent with the null result on net.

Russell 3000 — Monthly, Quarterly & Cap-Group Quarterly Temporal Figures



OBSERVATION

Russell 3000 quarterly and monthly series show more consistent positive coefficients than S&P 500, driven by mid - and small-cap segments. The 2022 and 2024 data gaps visible in the annual series are more pronounced at quarterly frequency.

Full Variable Definitions

Variable	Bloomberg Field	Definition	Panels
OUTCOMES			
return	derived	Open-to-close dollar return = px_close – px_open	Primary
return_oo	derived	Open-to-open return (%)	Extended, Russell
return_oo_adj	derived	Fama-French-Carhart 4-factor residual of return_oo	Extended, Russell
TREATMENTS			
twitter_sent	TWITTER_SENTIMENT_DAILY_AVG	Daily avg. Twitter sentiment $\in [-1,+1]$	All
twitter_neg_count	TWITTER_NEG_SENTIMENT_COUNT	Count of negative-polarity tweets	All
news_sent	NEWS_SENTIMENT_DAILY_AVG	Daily avg. news sentiment $\in [-1,+1]$	All
CONTROLS			
px_high / px_low	PX_HIGH / PX_LOW	Intraday high/low price (\$)	Primary
log_mkt_cap	derived	$\log(\text{CUR_MKT_CAP})$	All
vol_rs	derived	Rogers-Satchell realized volatility: $\sqrt{[(\ln h - \ln o)(\ln h - \ln c) + (\ln l - \ln o)(\ln l - \ln c)]}$	Ext., Russell
abnorm_vol	derived	% deviation of daily volume from ticker sample mean	Ext., Russell
rsi_30 / ma_50	RSI_30D / MOV_AVG_50D	30-day RSI; 50-day moving average price	Primary
bid_ask_spread	AVERAGE_BID_ASK_SPREAD	Daily avg. bid-ask spread (\$)	All
DERIVED			
lagk (k=1,2,3,5,7)	derived	return_{i,t-k} — lagged return for placebo/lag-structure tests	Primary
cap_group	derived	Large (>\$10B) / Mid (\$2B–\$10B) / Small (<\$2B) by avg. mkt_cap	Russell

DoubleML Partially Linear Regression — How It Works

The PLR Model (Chernozhukov et al. 2018)

Step 1 — Specify the PLR:

$$Y_{it} = \theta_0 \cdot D_{it} + g_0(X_{it}) + U_{it}$$

where $E[U_{it} \mid D_{it}, X_{it}] = 0$

$$D_{it} = m_0(X_{it}) + V_{it}$$

where $E[V_{it} \mid X_{it}] = 0$

Y_{it} = return (outcome)

D_{it} = twitter_sent (treatment)

X_{it} = confounder vector (controls)

$g_0(\cdot), m_0(\cdot)$ = unknown nonlinear functions $\rightarrow$ fit by XGBoost

θ_0 = causal parameter of interest

Step 2 — Cross-fit nuisance models (K=5 folds):

- Train $\hat{g}$ and $\hat{m}$ on 4 folds, predict on held-out fold
- Get residuals: $\hat{U} = Y - \hat{g}(X), \hat{V} = D - \hat{m}(X)$

Step 3 — Estimate θ by partialled-out OLS:

$$\theta = (\sum V_{it} \cdot \hat{U}_{it}) / (\sum V_{it} \cdot D_{it})$$

Step 4 — Repeat 20× over random fold splits $\rightarrow$ median θ

Step 5 — 1,000 bootstrap draws $\rightarrow$ SE and CI

Why DoubleML vs. OLS?

OLS assumes linear confounder–outcome and confounder–treatment relationships.

DoubleML relaxes this: g_0 and m_0 are fit nonparametrically — XGBoost captures interactions, nonlinearities, and collinear confounders automatically.

Neyman orthogonality: θ is $\sqrt{n}$ -consistent even when nuisance models converge at slower rates.

XGBoost as Nuisance Learner

Gradient-boosted tree ensemble; well-suited for:

- Mixed-type regressors (continuous + counts)
- Nonlinear and interaction effects via tree splits
- Robustness to multicollinearity among sentiment/momentum/volume
- Native handling of missing covariates

Identifying assumption: conditional unconfoundedness — no omitted variable jointly drives twitter_sent and return after conditioning on X_{it} .

Panel Fixed-Effects OLS — Estimating Equation

ESTIMATING EQUATION

$$Y_{it} = \alpha_i + \beta \cdot D_{it} + \gamma' \cdot X_{it} + \varepsilon_{it}$$

Notation	
Y_{it}	Open-to-close dollar return (return) for firm i on day t
α_i	Firm fixed effect — absorbs all time-invariant firm characteristics (industry, coverage, trading clientele)
D_{it}	Sentiment treatment: (1) twitter_sent (2) neg_twitter_sent (3) twitter_neg_count — estimated separately per model
β	Parameter of interest: within-firm partial effect of sentiment on same-day return
X_{it}	Control vector (M2): px_high, px_low, mkt_cap, total_equity, debt_to_equity, volume, news_sent, rsi_30, ma_50, twitter_neg_count
ε_{it}	Idiosyncratic error term — heteroskedasticity-robust SE (White sandwich estimator)

IDENTIFYING VARIATION

Within-firm, over-time variation only. Firm FE controls for stable differences across firms (size tier, analyst coverage, industry). The key assumption: unobserved confounders affect outcome linearly — relaxed by the DoubleML specification.

Standard errors: heteroskedasticity-robust (White sandwich). Fixed effects via within-transformation (demeaning). Software: pyfixest (Python).

Fama-MacBeth Procedure — Equation & Risk Adjustment

DAILY CROSS-SECTIONAL REGRESSION (per day t)

$$r_{it} = a_t + b_t \cdot \text{twit_sent}_{\{i,t-1\}} + c_t' \cdot X_{it} + e_{it}$$

FAMA-MACBETH AGGREGATION

$$\bar{b} = T^{-1} \cdot \sum_t b_t \quad [\text{Newey-West SE, 5 lags}]$$

FF4 RISK ADJUSTMENT

$$r_{oo_adj} = r_{oo} - (\beta^{MKT} \cdot MKT + \beta^{SMB} \cdot SMB + \beta^{HML} \cdot HML + \beta^{MOM} \cdot MOM)_t$$

Notation

r_{it}	Factor-adjusted open-to-open return (return_oo_adj) — not raw dollar return
$\text{twit_sent}_{\{i,t-1\}}$	One-day lagged Twitter sentiment (lagged to avoid contemporaneous endogeneity)
X_{it}	Controls: 5 lags each of return_oo_adj, abnorm_vol, vol_rs, log_mkt_cap, bid_ask_spread
$\bar{b}$	FMB estimate: time-series mean of daily cross-sectional slopes. t-stat from Newey-West SE
Exclusion rule	Days with fewer than 30 firms reporting non-missing twitter_sent are dropped from aggregation

TEMPORAL WINDOWS

Annual, quarterly, and monthly windows estimated separately. Annual used for tabular reporting; quarterly and monthly for visualization only. Cap-group analysis runs FMB independently within each cap tier.

Reference: Fama & MacBeth (1973), Gu & Kurov (2020). Factor data: Kenneth French Data Library.

Fama-French Factor Data — Kenneth French Data Library

Source & Access

mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Maintained by Kenneth R. French (Dartmouth Tuck). Free, publicly available. Updated monthly. US and international equities from 1926 to present.

Factors used in this study:

- MKT (Rm–Rf): Daily excess market return over risk-free rate (1-month T-bill)
- SMB: Small-Minus-Big — size factor (daily)
- HML: High-Minus-Low — value factor (daily)
- MOM: Up-Minus-Down — momentum factor (Carhart 1997, daily)

Frequency: Daily. Format: CSV. Truncated at Feb 28, 2026 (last available factor date at estimation time) to avoid extrapolation.

How Used in This Study

Step 1 — Estimate full-sample betas:

For each firm i , regress daily return_{oo} on MKT, SMB, HML, MOM over 2016–2026. Store β^{MKT} , β^{SMB} , β^{HML} , β^{MOM} per firm.

Step 2 — Compute factor-adjusted returns:

$$\text{return_oo_adj} = \text{return_oo} - (\beta^{\text{MKT}} \cdot \text{MKT}_t + \beta^{\text{SMB}} \cdot \text{SMB}_t + \beta^{\text{HML}} \cdot \text{HML}_t + \beta^{\text{MOM}} \cdot \text{MOM}_t)$$

Purpose: remove systematic risk exposures so FMB captures idiosyncratic sentiment effects only.

Step 3 — Use return_{oo_adj} as the dependent variable in all FMB specifications (S&P 500 extended and Russell 3000).

Reference: Carhart (1997) JF. Fama & French (1993) JFE.

NOTE

The FF4 model is the standard risk-adjustment in academic asset pricing studies. By using factor-adjusted returns, the FMB coefficient on twitter_sent is purged of variation attributable to market moves, size premiums, value premiums, and momentum — isolating idiosyncratic sentiment effects.

Bloomberg Sentiment Tool — Methodology & Academic Validation

How Bloomberg Classifies Twitter Sentiment

Step 1 — Human Expert Labeling (Gu & Kurov 2020):

Bloomberg lets human experts manually label tweets. Question: 'If an investor having a long position read this, would he/she be bullish, bearish, or neutral?'

Step 2 — Machine Learning Model Training:

Manually classified tweets feed into ML models trained to mimic expert labeling. Bloomberg does not disclose the specific model architecture or training data (proprietary).

Step 3 — Story-Level Scoring:

Each tweet assigned: score $S_k \in \{-1, 0, +1\}$ and confidence $C_k \in [0, 100\%]$.

Step 4 — Daily Aggregation Formula:

$$\text{Twitter}_{\{i,t\}} = [\sum P(i,T) \cdot S_k \cdot C_k] / N_{\{i,T\}}$$

$N_{\{i,T\}}$ = firm's total positive or negative tweet count in the 24-hour window $T = [t-24h, t]$.

Score released each morning ~10 minutes before market open.

Data licensing: Bloomberg signed a Twitter Data Licensing Agreement Sept. 2015 (Teti et al. 2019).

Academic Validation

Gu & Kurov (2020) — J. Banking & Finance:

First major peer-reviewed study using Bloomberg Twitter sentiment. Finds significant predictive power for Russell 3000 open-to-open returns.

Teti et al. (2019) — Technol. Forecasting & Social Change:

Bloomberg Twitter data for US tech sector. Finds significant relationship between `twitter_neg_count` and stock prices at lag 1.

Cam et al. (2024) — Heliyon:

ML classifiers for financial Twitter sentiment (SVM, MLP) achieve ~89% accuracy. Validates the ML approach underlying Bloomberg's system.

Loughran & McDonald (2011) critique:

General-purpose dictionaries misclassify financial language. Bloomberg's finance-specific supervised ML training directly addresses this — though the undisclosed architecture limits independent verification.

Long-Short Strategy & Sharpe Ratio — Construction and Results

Strategy Construction

Universe:

S&P 500 extended panel (2016–2026) and Russell 3000 cap groups.

Daily rebalancing:

Each day t , sort all firms with non-missing `twitter_sent` into deciles. Long the top decile (most positive sentiment), short the bottom decile (most negative sentiment). Equal-weighted within each leg.

Return measurement:

Daily long-short return = $\text{mean}(\text{top decile return}) - \text{mean}(\text{bottom decile return})$. Measured on `return_oo_adj` (FF4-adjusted open-to-open).

Sharpe Ratio (annualized):

$$SR = (\mu_{LS} - R_f) / \sigma_{LS} \times \sqrt{252}$$

μ_{LS} = mean daily long-short return

σ_{LS} = std dev of daily long-short returns

R_f = risk-free rate (daily 1-month T-bill; ≈ 0 in low-rate years)

$\sqrt{252}$ = annualization factor (trading days)

Reference: Sharpe (1966) 'Mutual Fund Performance,' *Journal of Business*.

Annual Long-Short Sharpe Ratios — S&P 500 Extended Panel

Year	L-S Return (%)	Sharpe Ratio	Signal
2016	positive	positive	✓
2017	positive	positive	✓
2018	positive	positive	✓
2019	positive	positive	✓
2020	positive	positive	✓
2021	positive	positive	✓
2022	positive	positive	✓
2023	top-decile	2.04	✓
2024	positive	positive	✓
2025	top-decile	1.61	✓

9 of 10 complete years show positive L-S returns.

THE REGRESSION VS. STRATEGY PUZZLE

FMB coefficient on `twitter_sent` is insignificant in 2023 and 2025 — yet the long-short Sharpe reaches 2.04 and 1.61. Explanation: FMB estimates the average cross-sectional slope (linear, across all firms). The long-short strategy exploits the tails of the sentiment distribution. If the sentiment–return relationship is nonlinear (strong in extremes, flat in the middle), the strategy profits while the linear regression finds a near-zero average slope. This suggests daily sentiment rankings — not levels — carry the exploitable signal.

Lagged DoubleML PLR — Sentiment Persistence Across Horizons (CATE by Lag)

Coefficient of lagged sentiment on current return (t)				
Lag	Twitter θ	95% CI	News θ	95% CI
t (base)	-0.924***	(-1.37, -0.48)	-0.759***	(-1.02, -0.50) ← lag 0
t-1	-1.215***	(-1.52, -0.91)	-1.169***	(-1.45, -0.89)
t-2	-0.957***	(-1.24, -0.67)	-0.793***	(-1.04, -0.54)
t-3	-1.122***	(-1.41, -0.83)	-1.363***	(-1.55, -1.18)
t-5	-1.126***	(-1.40, -0.86)	-0.943***	(-1.13, -0.76)
t-7	-0.674***	(-0.96, -0.39)	-0.638***	(-0.82, -0.46)

Each lag estimated separately. Outcome: return (day t). Confounders exclude contemporaneous sentiment. ***p<0.01. XGBoost, 5-fold, 20 reps, 1K bootstrap.

Coefficient magnitude by lag horizon



Time-Value of Twitter Sentiment — What the Lag Structure Reveals

Key Empirical Observations

OBSERVATION 1 — No monotonic decay

Under a clean price-discovery story, the effect of sentiment on return should shrink toward zero as the horizon increases — markets incorporate the signal quickly. That does not happen here. Coefficients at lag 5 (−1.13) and lag 7 (−0.67) remain large and highly significant. The signal does not appear to be 'used up.'

OBSERVATION 2 — Lag 1 exceeds lag 0

For both Twitter and news, the one-day-lagged coefficient (−1.22, −1.17) is larger in magnitude than the contemporaneous estimate (−0.92, −0.76). This is the opposite of what a forward-information channel predicts. It is, however, consistent with Bloomberg's 24-hour aggregation window: lag-1 sentiment partly reflects the prior day's price action, and that reverse channel compounds rather than attenuates across lags.

OBSERVATION 3 — Reverse causality persists across all horizons

The placebo test on the contemporaneous model found lag-1 return $\rightarrow$ twitter_sent = +2.348***. If sentiment reflects past prices and returns are autocorrelated (even weakly), lagged sentiment will mechanically co-move with current returns across multiple horizons. The lag decay pattern is more consistent with that mechanism than with sustained information production.

What This Means for Information Value

A genuine information channel would predict:

$$\theta(\text{lag } 0) > \theta(\text{lag } 1) > \theta(\text{lag } 2) > \dots \rightarrow 0$$

Signal is novel; once prices adjust, it no longer predicts.

What we observe instead:

$$\theta(\text{lag } 1) > \theta(\text{lag } 0) \text{ and } \theta(\text{lag } 5) \approx \theta(\text{lag } 1)$$

Signal strength does not diminish with time-to-incorporation.

Most consistent interpretation:

The Bloomberg daily score reflects prior return history at every horizon. Each lag captures the same underlying reverse-causality mechanism — not independent information arriving at each lag separately.

Implications

For identification:

Even with lagged treatment (the standard FMB approach), the coefficient does not cleanly identify a causal forward effect if the aggregation window creates carry-over from past returns into lagged sentiment.

For future research:

Timestamp-level tweet data would allow an instrument: pre-market sentiment (tweets posted 4am–9:30am ET) that cannot mechanically reflect intraday prices. This would sharpen the causal window at lag 0 and test whether a clean lag-1 effect survives.

For practitioners:

The long-short strategy earns positive returns despite these concerns — but that likely reflects return momentum embedded in the sentiment signal, not sentiment causally driving future prices.